

LPU 3D Campus Navigator

Product Requirements Document (PRD)

Purpose: Define the focused MVP for an interactive 3D campus navigation and discovery platform for Lovely Professional University.

1. Product Overview

A web application that represents the LPU campus as an interactive 3D environment. Users can explore buildings, discover vendors/services, ask verified campus questions through AI, and find the shortest practical walking route.

2. Core Problem

LPU is a large campus with many academic blocks, facilities, shops, food outlets and services. New students and visitors may struggle to locate places, understand what buildings contain, discover services, and determine practical routes.

Product goal: A user should be able to find any important place, understand it, and get there from one interface.

3. Target Users

New Student

Find classrooms, blocks, facilities, hostels and services quickly.

Existing Student

Navigate to unfamiliar locations and discover food, shops and services.

Visitor

Understand the campus and get simple directions.

Faculty/Staff

Reliably navigate to academic and administrative locations.

4. Strict MVP Feature List

F1 — Interactive 3D Campus Map

Rotate, zoom, pan, select buildings, show labels and reset camera.

F2 — Building/Location Information

Name, type, description, image, block identifier, coordinates, facilities and hours where applicable.

F3 — Search

Search buildings, blocks, shops, vendors, facilities and services; focus the map on the selected result.

F4 — Vendor & Shop Directory

Name, category, location, description, hours and contact information where appropriate.

F5 — Shortest Walking Route

Select start and destination; calculate a route using a campus graph. MVP algorithm: Dijkstra. Show route, distance and approximate walking time.

F6 — AI Campus Chatbot

Answer campus questions using verified application data. Never invent information.

F7 — Basic Mobile Experience

Usable on mobile, tablet and desktop.

5. MVP Non-Goals

- AR navigation
- Indoor turn-by-turn navigation
- Real-time crowd tracking
- Live vendor inventory
- Payments
- Social networking
- User-generated reviews
- Complex user profiles
- Campus event management
- Native Android/iOS apps

6. Success Metrics

- Percentage of users successfully finding a destination
- Search → location selection success rate
- Route calculation success rate
- Average time required to find a location
- Chatbot successful-answer rate
- Location exploration and vendor interactions

7. Primary User Flow

Open app → 3D map → Search/Explore → Select location → Information panel → Navigate → Start + Destination → Shortest route → Route summary